

B2 Symmetry and Relativity — Model Solutions, 2017

Question 1 — 4-vectors, pure force, annihilation

Problem. Define $\mathbf{U}$, $\mathbf{F}$; show orthogonal non-zero 4-vectors split into space-/time-like; show $\mathbf{P} \cdot \mathbf{P} = 0 \Rightarrow m = 0$. Define proper time, pure force; show $\mathbf{F} \cdot \mathbf{U} = 0$ and (with $\mathbf{v} \perp \mathbf{a}$) $A^\mu A_\mu = \gamma^4 a^2$. Solve constant pure-force motion; treat e^+e^- annihilation; find γ, β for an e^- accelerated through $L = 3\text{ m}$ in $E = 10\text{ MV/m}$.

(a) Definitions and 4-orthogonality

4-velocity and 4-force:

$$U^\mu = \frac{dX^\mu}{d\tau} = \gamma(c, \mathbf{v}), \quad F^\mu = \frac{dP^\mu}{d\tau}, \quad P^\mu = mU^\mu.$$

Orthogonality $\mathbf{D} \cdot \mathbf{G} = 0$ with both non-zero. Assume $\mathbf{D}$ is time-like and boost to its rest frame:

$$\mathbf{D} = (D^0, \mathbf{0}), \quad D^0 \neq 0.$$

Expand the inner product in this frame:

$$\mathbf{D} \cdot \mathbf{G} = D^0 G^0 = 0.$$

Divide by $D^0 \neq 0$:

$$G^0 = 0.$$

Compute the invariant norm of $\mathbf{G}$:

$$\mathbf{G} \cdot \mathbf{G} = (G^0)^2 - |\mathbf{G}|^2 = -|\mathbf{G}|^2.$$

Use $\mathbf{G} \neq 0$ with $G^0 = 0$, so $\mathbf{G} \neq \mathbf{0}$:

$$\mathbf{G} \cdot \mathbf{G} < 0.$$

Hence $\mathbf{G}$ is space-like whenever $\mathbf{D}$ is time-like. Symmetrically, if $\mathbf{G}$ is time-like then $\mathbf{D}$ is space-like. Two time-like vectors cannot be orthogonal: in the rest frame of one, the other still has non-zero time component (a time-like 4-vector has $G^0 \neq 0$ in every frame), so $D^0 G^0 \neq 0$. Therefore exactly one is time-like and the other space-like. $\square$

Invariant norm of 4-momentum:

$$\mathbf{P} \cdot \mathbf{P} = \frac{E^2}{c^2} - |\mathbf{p}|^2 = m^2 c^2.$$

Set $\mathbf{P} \cdot \mathbf{P} = 0$:

$$m^2 c^2 = 0.$$

Solve for m :

$$\boxed{m = 0}.$$

Example: photon.

(b) Proper time, pure force, $A^\mu A_\mu = \gamma^4 a^2$

Proper time: time interval recorded by a clock comoving with the particle, $d\tau = dt/\gamma$. **Pure force:** a 4-force that does not change rest mass, $dm/d\tau = 0$.

Start from $\mathbf{P} \cdot \mathbf{P} = m^2 c^2$:

$$P^\mu P_\mu = m^2 c^2.$$

Differentiate w.r.t. τ :

$$2P^\mu \frac{dP_\mu}{d\tau} = 2mc^2 \frac{dm}{d\tau}.$$

Identify $dP^\mu/d\tau = F^\mu$:

$$P^\mu F_\mu = mc^2 \frac{dm}{d\tau}.$$

Pure force gives $dm/d\tau = 0$:

$$P^\mu F_\mu = 0.$$

Divide by m using $P^\mu = mU^\mu$:

$$\boxed{\mathbf{F} \cdot \mathbf{U} = 0}.$$

4-acceleration: $A^\mu = dU^\mu/d\tau$. Convert proper-time to lab-time via $d\tau = dt/\gamma$:

$$A^\mu = \gamma \frac{d}{dt}(\gamma c, \gamma \mathbf{v}).$$

Expand the derivative component by component:

$$A^\mu = (\gamma \dot{\gamma} c, \gamma \dot{\gamma} \mathbf{v} + \gamma^2 \mathbf{a}).$$

Use the standard identity $\dot{\gamma} = \gamma^3(\mathbf{v} \cdot \mathbf{a})/c^2$:

$$\dot{\gamma} = \gamma^3(\mathbf{v} \cdot \mathbf{a})/c^2.$$

Apply $\mathbf{v} \perp \mathbf{a}$, so $\mathbf{v} \cdot \mathbf{a} = 0$:

$$\dot{\gamma} = 0.$$

Substitute back:

$$A^\mu = (0, \gamma^2 \mathbf{a}).$$

Square in $(+, -, -, -)$ signature:

$$A^\mu A_\mu = (A^0)^2 - |\mathbf{A}|^2 = 0 - \gamma^4 a^2.$$

Take the magnitude:

$$\boxed{|A^\mu A_\mu| = \gamma^4 a^2}.$$

(c) Constant pure force on rest mass M_0

Newton's law in lab time with constant 3-force f_x :

$$\frac{dp_x}{dt} = f_x.$$

Integrate with initial condition $p_x(0) = 0$:

$$p_x(t) = f_x t.$$

Energy-momentum relation:

$$E^2 = (M_0 c^2)^2 + (p_x c)^2.$$

Substitute $p_x = f_x t$:

$$E(t)^2 = (M_0 c^2)^2 + (f_x t c)^2.$$

Take the square root:

$$E(t) = c \sqrt{(M_0 c)^2 + (f_x t)^2}.$$

Use $U^0 = E/(M_0 c)$:

$$U^0 = \frac{c \sqrt{(M_0 c)^2 + (f_x t)^2}}{M_0 c}.$$

Factor $M_0 c$ inside the root:

$$U^0 = c \sqrt{1 + (f_x t / (M_0 c))^2}.$$

Use $U^1 = p_x / M_0$:

$$U^1 = \frac{f_x t}{M_0}.$$

Assemble U^μ :

$$U^\mu = \left(c \sqrt{1 + (f_x t / (M_0 c))^2}, \frac{f_x t}{M_0}, 0, 0 \right).$$

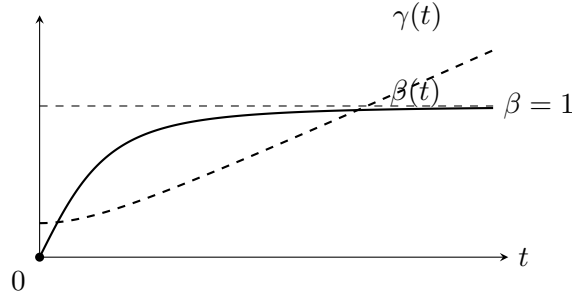
Read off $\gamma = U^0 / c$:

$$\gamma(t) = \sqrt{1 + (f_x t / (M_0 c))^2}.$$

Use $\beta = p_x c / E$:

$$\beta(t) = \frac{f_x t}{\sqrt{(M_0 c)^2 + (f_x t)^2}}.$$

Limits: $\beta(0) = 0$, $\gamma(0) = 1$; $\beta \rightarrow 1$ and $\gamma \rightarrow f_x t / (M_0 c)$ as $t \rightarrow \infty$.



(d) $e^+ e^-$ annihilation at rest — minimum photon count

Initial system has zero net 3-momentum (head-on, equal speeds, equal mass). A single photon would carry $|\mathbf{p}| = E_\gamma / c \neq 0$, violating momentum conservation. Hence

$$N_{\min} = 2 \quad (\text{back-to-back, equal energy}).$$

Energy conservation in lab frame:

$$2E_\gamma = (T_{e^-} + m_e c^2) + (T_{e^+} + m_e c^2).$$

Group kinetic and rest-energy terms:

$$2E_\gamma = T_{\text{tot}} + 2m_e c^2.$$

Substitute $T_{\text{tot}} = 1000 \text{ MeV}$, $m_e c^2 = 0.511 \text{ MeV}$:

$$2E_\gamma = 1000 + 2(0.511).$$

Evaluate the right-hand side:

$$2E_\gamma = 1001.022 \text{ MeV}.$$

Divide by 2:

$$\boxed{E_\gamma \approx 500.5 \text{ MeV}}.$$

(e) Electron through $L = 3 \text{ m}$ at $E = 10 \text{ MV/m}$

Kinetic energy gained equals work done by the field:

$$T = eEL.$$

Substitute $eE = 10 \text{ MeV/m}$, $L = 3 \text{ m}$:

$$T = (10 \text{ MeV/m}) \times (3 \text{ m}).$$

Evaluate:

$$T = 30 \text{ MeV}.$$

Compute total energy $W = T + m_e c^2$:

$$W = 30 + 0.511 \text{ MeV} = 30.511 \text{ MeV}.$$

Lorentz factor $\gamma = W/(m_e c^2)$:

$$\gamma = \frac{30.511}{0.511}.$$

Evaluate:

$$\boxed{\gamma \approx 59.7}.$$

Compute $1/\gamma^2$:

$$1/\gamma^2 \approx 2.81 \times 10^{-4}.$$

Apply $\beta = \sqrt{1 - 1/\gamma^2}$:

$$\beta = \sqrt{1 - 2.81 \times 10^{-4}}.$$

Evaluate:

$$\boxed{\beta \approx 0.99986}.$$

Question 2 — 4-wavevector, simultaneity, fields, photon-gas invariant

Problem. Components of $\mathbf{K}$; phase invariance; isolated electron cannot emit a photon; space-like condition and simultaneity frame; two-photon system invariant mass and CoM velocity; non-relativistic motion in crossed $\mathbf{E}, \mathbf{B}$ for $|E/B| \gtrless c$; invariance of $s^2 = (\sum W)^2 - |\sum \mathbf{p}|^2$ and its value for a photon gas.

(a) 4-wavevector, phase invariance, no spontaneous photon emission

For a plane EM wave $\psi \propto e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ in vacuum with $|\mathbf{k}| = \omega/c$:

$$K^\mu = (\omega/c, \mathbf{k}).$$

Form the contraction $K_\mu X^\mu$ with $X^\mu = (ct, \mathbf{x})$:

$$K_\mu X^\mu = (\omega/c)(ct) - \mathbf{k} \cdot \mathbf{x} = \omega t - \mathbf{k} \cdot \mathbf{x}.$$

Identify the wave phase:

$$\varphi = \mathbf{k} \cdot \mathbf{x} - \omega t = -K_\mu X^\mu.$$

Contraction of two 4-vectors is a Lorentz scalar:

$$\boxed{\varphi' = \varphi}.$$

Hypothetical process $e^- \rightarrow e^- + \gamma$ in vacuum. Work in the initial electron rest frame:

$$\mathbf{P}_e = (m_e c, \mathbf{0}).$$

4-momentum conservation:

$$\mathbf{P}_e = \mathbf{P}'_e + \hbar \mathbf{K}_\gamma.$$

Read off 3-momentum: $\mathbf{p}'_e = -\hbar \mathbf{k}_\gamma$:

$$|\mathbf{p}'_e| = \hbar |\mathbf{k}_\gamma| = \hbar \omega / c.$$

Apply energy-momentum relation to the final electron:

$$E'_e = \sqrt{(m_e c^2)^2 + |\mathbf{p}'_e|^2 c^2} = \sqrt{(m_e c^2)^2 + (\hbar \omega)^2}.$$

Apply energy conservation:

$$m_e c^2 = \sqrt{(m_e c^2)^2 + (\hbar \omega)^2} + \hbar \omega.$$

Isolate the square root:

$$m_e c^2 - \hbar \omega = \sqrt{(m_e c^2)^2 + (\hbar \omega)^2}.$$

Square both sides:

$$(m_e c^2)^2 - 2m_e c^2 \hbar \omega + (\hbar \omega)^2 = (m_e c^2)^2 + (\hbar \omega)^2.$$

Cancel $(m_e c^2)^2$ and $(\hbar \omega)^2$:

$$-2m_e c^2 \hbar \omega = 0.$$

Solve for ω :

$$\omega = 0.$$

No physical photon emission possible. □

(b) Space-like interval, simultaneity

Invariant interval between $\mathbf{D}$ and $\mathbf{G}$:

$$(\mathbf{D} - \mathbf{G}) \cdot (\mathbf{D} - \mathbf{G}) = c^2(t_d - t_g)^2 - |\mathbf{x}_d - \mathbf{x}_g|^2.$$

Space-like condition (interval < 0):

$$\boxed{|\mathbf{x}_d - \mathbf{x}_g|^2 > c^2(t_d - t_g)^2}.$$

Boost with velocity $\mathbf{V}$ parallel to $\Delta\mathbf{x} = \mathbf{x}_d - \mathbf{x}_g$:

$$\Delta t' = \gamma_V (\Delta t - \mathbf{V} \cdot \Delta\mathbf{x}/c^2).$$

Set $\Delta t' = 0$ (simultaneity in S'):

$$\Delta t = \mathbf{V} \cdot \Delta\mathbf{x}/c^2.$$

Use $\mathbf{V} \parallel \Delta\mathbf{x}$, so $\mathbf{V} \cdot \Delta\mathbf{x} = V|\Delta\mathbf{x}|$:

$$V = \frac{c^2 \Delta t}{|\Delta\mathbf{x}|}.$$

Compute $|V|/c$:

$$|V|/c = \frac{c|\Delta t|}{|\Delta\mathbf{x}|}.$$

Apply the space-like condition $c|\Delta t| < |\Delta\mathbf{x}|$:

$$|V|/c < 1.$$

Hence a real boost exists — such a frame S' exists.

(c) Two perpendicular photons of equal ω

4-momenta of two photons (components $(E/c, \mathbf{p})$) with $|\mathbf{p}| = \hbar\omega/c$:

$$\mathbf{P}_1 = \frac{\hbar\omega}{c}(1, 1, 0, 0), \quad \mathbf{P}_2 = \frac{\hbar\omega}{c}(1, 0, 1, 0).$$

Add the 4-momenta:

$$\mathbf{P} = \mathbf{P}_1 + \mathbf{P}_2 = \frac{\hbar\omega}{c}(2, 1, 1, 0).$$

Form the invariant norm in $(+, -, -, -)$:

$$\mathbf{P} \cdot \mathbf{P} = \left(\frac{\hbar\omega}{c}\right)^2 (2^2 - 1^2 - 1^2).$$

Evaluate:

$$\mathbf{P} \cdot \mathbf{P} = 2\left(\frac{\hbar\omega}{c}\right)^2.$$

Identify $\mathbf{P} \cdot \mathbf{P} = M^2 c^2$:

$$M^2 c^2 = 2\left(\frac{\hbar\omega}{c}\right)^2.$$

Solve for M :

$$\boxed{M = \frac{\sqrt{2} \hbar\omega}{c^2}}.$$

CoM velocity formula: $\mathbf{v}_{\text{CoM}} = c^2 \mathbf{p}_{\text{tot}} / E_{\text{tot}}$. Read off totals:

$$\mathbf{p}_{\text{tot}} = \frac{\hbar\omega}{c}(1, 1, 0), \quad E_{\text{tot}} = 2\hbar\omega.$$

Substitute:

$$\mathbf{v}_{\text{CoM}} = \frac{c^2(\hbar\omega/c)(1, 1, 0)}{2\hbar\omega}.$$

Cancel $\hbar\omega$:

$$\mathbf{v}_{\text{CoM}} = \frac{c}{2}(1, 1, 0).$$

Compute the magnitude:

$$|\mathbf{v}_{\text{CoM}}| = \frac{c}{2}\sqrt{1^2 + 1^2} = \frac{c}{\sqrt{2}}.$$

Final answer:

$$\boxed{\mathbf{v}_{\text{CoM}} = (c/2, c/2, 0), \quad |\mathbf{v}_{\text{CoM}}| = c/\sqrt{2}}.$$

(d) Electron in crossed $\mathbf{E} \perp \mathbf{B}$

Field invariants:

$$\mathbf{E} \cdot \mathbf{B} = 0, \quad I_2 = E^2 - c^2 B^2.$$

Compute $\mathbf{E} \times \mathbf{B}$ with $\mathbf{E} = (0, E, 0)$, $\mathbf{B} = (0, 0, B)$:

$$\mathbf{E} \times \mathbf{B} = E\hat{\mathbf{y}} \times B\hat{\mathbf{z}} = EB\hat{\mathbf{x}}.$$

Case (a): $|E/B| > c$, so $I_2 > 0$. Choose boost along $\hat{\mathbf{x}}$ that eliminates $\mathbf{B}'$:

$$\mathbf{v}_d = \frac{c^2 \mathbf{E} \times \mathbf{B}}{E^2} = \left(\frac{c^2 B}{E}, 0, 0 \right).$$

Check $|\mathbf{v}_d|/c < 1$:

$$|\mathbf{v}_d|/c = cB/E < 1 \quad (\text{since } E/B > c).$$

Define $\gamma_d = 1/\sqrt{1 - c^2 B^2/E^2}$. Compute E'_y :

$$E'_y = \gamma_d(E_y - v_d B_z) = \gamma_d(E - (c^2 B/E) B).$$

Factor E :

$$E'_y = \gamma_d E (1 - c^2 B^2/E^2).$$

Use $1 - c^2 B^2/E^2 = 1/\gamma_d^2$:

$$E'_y = E/\gamma_d.$$

Compute B'_z :

$$B'_z = \gamma_d(B_z - v_d E_y/c^2) = \gamma_d(B - (c^2 B/E)(E)/c^2).$$

Cancel inside the bracket:

$$B'_z = 0.$$

Hence $\mathbf{B}' = 0$ and $\mathbf{E}' = (E/\gamma_d)\hat{\mathbf{y}}$. Equation of motion in S' for electron of charge $-e$:

$$\frac{d\mathbf{p}'}{dt'} = -e\mathbf{E}'.$$

Initial condition: electron at rest in S , so in S' it moves at $-\mathbf{v}_d \hat{\mathbf{x}}'$. Result: uniform acceleration along $-\hat{\mathbf{y}}'$ superposed on $-\mathbf{v}_d \hat{\mathbf{x}}'$, giving relativistic hyperbolic motion in S' (no closed orbit). In lab frame S : add drift $+\mathbf{v}_d \hat{\mathbf{x}}$ — particle is continuously accelerated and drifts in $+\hat{\mathbf{x}}$, an open unbounded trajectory.

Case (b): $|E/B| < c$, so $I_2 < 0$. Choose boost along $\hat{\mathbf{x}}$ that eliminates $\mathbf{E}'$:

$$\mathbf{v}_d = \frac{\mathbf{E} \times \mathbf{B}}{B^2} = \left(\frac{E}{B}, 0, 0 \right), \quad |\mathbf{v}_d| < c.$$

Define $\gamma_d = 1/\sqrt{1 - E^2/(c^2 B^2)}$. Compute E'_y :

$$E'_y = \gamma_d(E_y - v_d B_z) = \gamma_d(E - (E/B)B).$$

Cancel inside the bracket:

$$E'_y = 0.$$

Compute B'_z :

$$B'_z = \gamma_d(B_z - v_d E_y/c^2) = \gamma_d B(1 - E^2/(c^2 B^2)).$$

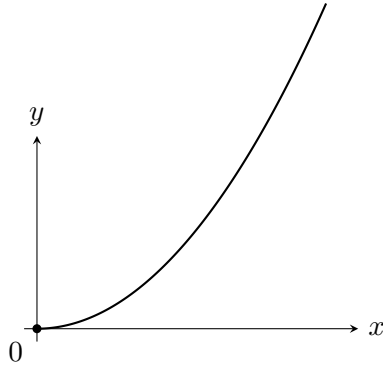
Use $1 - E^2/(c^2 B^2) = 1/\gamma_d^2$:

$$B'_z = B/\gamma_d.$$

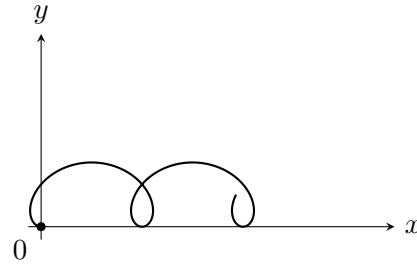
Hence $\mathbf{E}' = 0$ and $\mathbf{B}' = (B/\gamma_d)\hat{\mathbf{z}}$. Equation of motion in S' :

$$\frac{d\mathbf{p}'}{dt'} = -e \mathbf{v}' \times \mathbf{B}'.$$

Initial condition: in S' electron moves at $-\mathbf{v}_d \hat{\mathbf{x}}'$. Pure magnetic field gives a closed circular orbit in the $x'y'$ -plane of radius $r' = p'/(eB')$. In lab frame S : add drift $+\mathbf{v}_d \hat{\mathbf{x}}$ — cycloid drifting along $+\hat{\mathbf{x}}$ at speed E/B .



Case (a): $|E/B| > c$



Case (b): $|E/B| < c$ (cycloid)

(e) Invariance of s^2 for non-interacting system

Define total 4-momentum:

$$\mathbf{P}_{\text{tot}} = \sum_i \mathbf{P}_i.$$

Sum of 4-vectors is a 4-vector; non-interacting particles have each $\mathbf{P}_i$ conserved. Form the invariant norm (with $c = 1$):

$$\mathbf{P}_{\text{tot}} \cdot \mathbf{P}_{\text{tot}} = \left(\sum_i W_i \right)^2 - \left| \sum_i \mathbf{p}_i \right|^2 = s^2.$$

Inner product of a 4-vector with itself is a Lorentz scalar, so s^2 is invariant. $\square$

Photon gas: each photon satisfies $\mathbf{P}_i \cdot \mathbf{P}_i = W_i^2 - |\mathbf{p}_i|^2 = 0$, so $|\mathbf{p}_i| = W_i$. Expand the squared norm:

$$s^2 = \sum_i \mathbf{P}_i \cdot \mathbf{P}_i + 2 \sum_{i < j} \mathbf{P}_i \cdot \mathbf{P}_j.$$

Drop the null self-terms:

$$s^2 = 2 \sum_{i < j} \mathbf{P}_i \cdot \mathbf{P}_j.$$

Expand each inner product:

$$\mathbf{P}_i \cdot \mathbf{P}_j = W_i W_j - \mathbf{p}_i \cdot \mathbf{p}_j.$$

Substitute $\mathbf{p}_i \cdot \mathbf{p}_j = |\mathbf{p}_i| |\mathbf{p}_j| \cos \theta_{ij} = W_i W_j \cos \theta_{ij}$:

$$\mathbf{P}_i \cdot \mathbf{P}_j = W_i W_j (1 - \cos \theta_{ij}).$$

Sum over pairs:

$$s^2 = 2 \sum_{i < j} W_i W_j (1 - \cos \theta_{ij}) \geq 0.$$

Question 3 — Astrophysical SR tests, Doppler, elastic collision, undulator

Problem. Three astrophysical SR tests; Compton shift; relativistic Doppler longitudinal and transverse; equal-mass elastic-collision angle theorem; field of a static undulator in the electron rest frame.

(a) Three astrophysical tests; Compton shift

Three direct SR tests via emitted/reflected light:

1. Cosmological redshift of distant galaxies (relativistic Doppler).
2. Time dilation of supernova light curves (Type-Ia at high z).
3. Aberration of starlight / transverse Doppler from rapidly moving stellar systems (or relativistic beaming in jets).

Compton scattering: incoming photon $\hbar \mathbf{K} = (\hbar \omega / c, \hbar \mathbf{k})$, electron at rest $\mathbf{P}_e = (m_e c, \mathbf{0})$. Outgoing: photon $\hbar \mathbf{K}' = (\hbar \omega' / c, \hbar \mathbf{k}')$, electron $\mathbf{P}'_e$ with $\mathbf{P}'_e \cdot \mathbf{P}'_e = m_e^2 c^2$.

Apply 4-momentum conservation:

$$\hbar \mathbf{K} + \mathbf{P}_e = \hbar \mathbf{K}' + \mathbf{P}'_e.$$

Isolate the scattered electron:

$$\mathbf{P}'_e = \hbar \mathbf{K} - \hbar \mathbf{K}' + \mathbf{P}_e.$$

Square both sides (take invariant norm):

$$\mathbf{P}'_e \cdot \mathbf{P}'_e = (\hbar \mathbf{K} - \hbar \mathbf{K}' + \mathbf{P}_e) \cdot (\hbar \mathbf{K} - \hbar \mathbf{K}' + \mathbf{P}_e).$$

Expand the right-hand side:

$$m_e^2 c^2 = \hbar^2 \mathbf{K} \cdot \mathbf{K} + \hbar^2 \mathbf{K}' \cdot \mathbf{K}' - 2 \hbar^2 \mathbf{K} \cdot \mathbf{K}' + 2 \hbar \mathbf{P}_e \cdot \mathbf{K} - 2 \hbar \mathbf{P}_e \cdot \mathbf{K}' + m_e^2 c^2.$$

Apply $\mathbf{K} \cdot \mathbf{K} = \mathbf{K}' \cdot \mathbf{K}' = 0$ (massless photons):

$$0 = -2\hbar^2 \mathbf{K} \cdot \mathbf{K}' + 2\hbar \mathbf{P}_e \cdot \mathbf{K} - 2\hbar \mathbf{P}_e \cdot \mathbf{K}'.$$

Divide by $2\hbar$:

$$\mathbf{P}_e \cdot \mathbf{K} - \mathbf{P}_e \cdot \mathbf{K}' = \hbar \mathbf{K} \cdot \mathbf{K}'.$$

Compute $\mathbf{P}_e \cdot \mathbf{K}$ with $\mathbf{P}_e = (m_e c, \mathbf{0})$:

$$\mathbf{P}_e \cdot \mathbf{K} = (m_e c)(\omega/c) - \mathbf{0} \cdot \hbar \mathbf{k} = m_e \omega.$$

Similarly:

$$\mathbf{P}_e \cdot \mathbf{K}' = m_e \omega'.$$

Compute $\mathbf{K} \cdot \mathbf{K}'$ with $|\mathbf{k}| = \omega/c$, $|\mathbf{k}'| = \omega'/c$:

$$\mathbf{K} \cdot \mathbf{K}' = \frac{\omega \omega'}{c^2} - \mathbf{k} \cdot \mathbf{k}'.$$

Substitute $\mathbf{k} \cdot \mathbf{k}' = (\omega \omega' / c^2) \cos \theta$:

$$\mathbf{K} \cdot \mathbf{K}' = \frac{\omega \omega'}{c^2} (1 - \cos \theta).$$

Substitute into $\mathbf{P}_e \cdot \mathbf{K} - \mathbf{P}_e \cdot \mathbf{K}' = \hbar \mathbf{K} \cdot \mathbf{K}'$:

$$m_e(\omega - \omega') = \frac{\hbar \omega \omega'}{c^2} (1 - \cos \theta).$$

Divide by $m_e \omega \omega'$:

$$\frac{1}{\omega'} - \frac{1}{\omega} = \frac{\hbar}{m_e c^2} (1 - \cos \theta).$$

Final Compton formula:

$$\boxed{\frac{1}{\omega'} - \frac{1}{\omega} = \frac{\hbar}{m_e c^2} (1 - \cos \theta)} \iff \Delta \lambda = \frac{h}{m_e c} (1 - \cos \theta).$$

(b) Doppler effect

Phase $\varphi = -K_\mu X^\mu$ is Lorentz invariant; equivalently $\mathbf{K}$ transforms as a 4-vector. Vacuum dispersion gives $|\mathbf{k}| = \omega/c$, so with $\mathbf{k} = k_x \hat{\mathbf{x}}$:

$$\mathbf{K} = (\omega/c, \omega/c, 0, 0).$$

Case (a): $\mathbf{v} = (v_x, 0, 0)$ parallel to $\mathbf{k}$ (longitudinal). Apply Lorentz boost along $\hat{\mathbf{x}}$ to K^0 :

$$K'^0 = \gamma(K^0 - \beta K^1).$$

Substitute $K^0 = K^1 = \omega/c$:

$$\omega'/c = \gamma(\omega/c - \beta \omega/c).$$

Factor ω/c :

$$\omega' = \gamma \omega (1 - \beta).$$

Use $\gamma = 1/\sqrt{(1 - \beta)(1 + \beta)}$:

$$\omega' = \omega \frac{1 - \beta}{\sqrt{(1 - \beta)(1 + \beta)}}.$$

Simplify the fraction:

$$\boxed{\omega' = \omega \sqrt{\frac{1-\beta}{1+\beta}}} \quad (\text{longitudinal Doppler}).$$

Case (b): $\mathbf{v} = (0, v_y, 0)$ **perpendicular to $\mathbf{k}$ (transverse).** Apply Lorentz boost along $\hat{\mathbf{y}}$ to K^0 :

$$K'^0 = \gamma(K^0 - \beta K^2).$$

Substitute $K^2 = 0$:

$$\omega'/c = \gamma\omega/c.$$

Multiply by c :

$$\boxed{\omega' = \gamma\omega} \quad (\text{transverse Doppler}).$$

(c) Elastic collision: equal post-collision angles $\Rightarrow M_1 = M_2$

Work in the CoM frame. By definition, total 3-momentum vanishes, so outgoing particles are back-to-back with equal momentum magnitude:

$$|\mathbf{p}_1^*| = |\mathbf{p}_2^*| \equiv p^*.$$

In elastic scattering CoM energies are conserved and fixed by the masses:

$$E_i^* = \sqrt{p^{*2}c^2 + M_i^2c^4} \quad (i = 1, 2).$$

Let particle 1 emerge at angle ϕ^* above $+\hat{\mathbf{x}}$ in CoM; particle 2 at angle $\pi + \phi^*$:

$$p_{1x}^* = p^* \cos \phi^*, \quad p_{1y}^* = p^* \sin \phi^*; \quad p_{2x}^* = -p^* \cos \phi^*, \quad p_{2y}^* = -p^* \sin \phi^*.$$

Boost back to lab along $+\beta_{\text{CoM}}\hat{\mathbf{x}}$. Transverse momenta unchanged:

$$p_{1y}^{\text{lab}} = p^* \sin \phi^*, \quad p_{2y}^{\text{lab}} = -p^* \sin \phi^*.$$

Transform longitudinal components:

$$p_{1x}^{\text{lab}} = \gamma_{\text{CoM}}(p^* \cos \phi^* + \beta_{\text{CoM}}E_1^*/c),$$

$$p_{2x}^{\text{lab}} = \gamma_{\text{CoM}}(-p^* \cos \phi^* + \beta_{\text{CoM}}E_2^*/c).$$

Define lab angles θ_i (above/below $\hat{\mathbf{x}}$) by $\tan \theta_i = |p_{iy}^{\text{lab}}|/p_{ix}^{\text{lab}}$:

$$\tan \theta_1 = \frac{p^* \sin \phi^*}{\gamma_{\text{CoM}}(p^* \cos \phi^* + \beta_{\text{CoM}}E_1^*/c)},$$

$$\tan \theta_2 = \frac{p^* \sin \phi^*}{\gamma_{\text{CoM}}(-p^* \cos \phi^* + \beta_{\text{CoM}}E_2^*/c)}.$$

Impose $\theta_1 = \theta_2$ (numerators equal, so denominators must be equal):

$$p^* \cos \phi^* + \beta_{\text{CoM}}E_1^*/c = -p^* \cos \phi^* + \beta_{\text{CoM}}E_2^*/c.$$

Rearrange:

$$2p^* \cos \phi^* = \beta_{\text{CoM}}(E_2^* - E_1^*)/c.$$

The CoM scattering angle ϕ^* depends on impact parameter and can take any value in $[0, \pi]$, while E_1^* , E_2^* , p^* , β_{CoM} are fixed by initial kinematics. For $\theta_1 = \theta_2$ to hold for the elastic scattering (independently of the impact parameter, i.e., for all ϕ^*), the right-hand side must be independent of ϕ^* and equal the variable left-hand side; both sides must therefore vanish:

$$E_1^* = E_2^* \quad \text{and} \quad \cos \phi^* = 0.$$

Apply $|\mathbf{p}_1^*| = |\mathbf{p}_2^*| = p^*$ and the energy-momentum relation $E_i^{*2} - p^{*2}c^2 = M_i^2c^4$:

$$M_1^2c^4 = (E_1^*)^2 - p^{*2}c^2 = (E_2^*)^2 - p^{*2}c^2 = M_2^2c^4.$$

Take the positive square root:

$$\boxed{M_1 = M_2}. \quad \square$$

(d) Static undulator field in electron rest frame

Lab potentials: $\mathbf{A} = (0, A_0 \cos(k_u z), 0)$, $\phi = 0$. Compute the lab electric field:

$$\mathbf{E} = -\partial_t \mathbf{A} - \nabla \phi = \mathbf{0}.$$

Compute the lab magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$:

$$B_x = \partial_y A_z - \partial_z A_y = -\partial_z [A_0 \cos(k_u z)].$$

Differentiate:

$$B_x = A_0 k_u \sin(k_u z).$$

Other components vanish:

$$\mathbf{B} = A_0 k_u \sin(k_u z) \hat{\mathbf{x}}.$$

Boost to electron rest frame along $+\hat{\mathbf{z}}$ with velocity $v\hat{\mathbf{z}}$, $\gamma = 1/\sqrt{1 - v^2/c^2}$. Apply field transformation $\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B})$ with $\mathbf{E} = \mathbf{0}$:

$$\mathbf{v} \times \mathbf{B} = v\hat{\mathbf{z}} \times B_x \hat{\mathbf{x}} = vB_x \hat{\mathbf{y}}.$$

Read off E'_y :

$$E'_y = \gamma v B_x = \gamma v A_0 k_u \sin(k_u z).$$

Other electric components:

$$E'_x = 0, \quad E'_z = E_z = 0.$$

Apply magnetic transformation $\mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \mathbf{v} \times \mathbf{E}/c^2)$ with $\mathbf{E} = \mathbf{0}$:

$$B'_x = \gamma B_x = \gamma A_0 k_u \sin(k_u z).$$

Other magnetic components:

$$B'_y = 0, \quad B'_z = B_z = 0.$$

Express z in primed coordinates via inverse Lorentz transformation $z = \gamma(z' + vt')$:

$$\boxed{\mathbf{E}' = \gamma v A_0 k_u \sin(k_u \gamma(z' + vt')) \hat{\mathbf{y}}, \quad \mathbf{B}' = \gamma A_0 k_u \sin(k_u \gamma(z' + vt')) \hat{\mathbf{x}}}.$$

Wave-like check. Compute ratio $|E'|/|B'|$:

$$|E'|/|B'| = v.$$

Read off wave-vector and frequency from the phase $k_u \gamma(z' + vt')$:

$$k' = \gamma k_u, \quad \omega' = \gamma v k_u.$$

Compute phase velocity:

$$\omega'/k' = v < c.$$

Strictly $\omega'/k' \neq c$, so it is *not* a vacuum EM wave; it is the Lorentz-transformed near-field of a static magnet. In the ultra-relativistic limit $v \rightarrow c$ it becomes indistinguishable from a propagating EM wave (basis of free-electron-laser/undulator radiation).

Question 4 — Field transformations, EM tensor, wires, inverse Compton

Problem. Field transformations under boost along x ; field tensor from a plane-wave 4-potential; Maxwell in covariant form; physical meaning of f_0 ; orthogonality of U_W^μ and $\mathbf{P}$; fields of a current-carrying wire in three frames and whether two wires can interact purely electrically; ultra-relativistic inverse Compton maximum.

(a) Field transformations, boost along $\hat{\mathbf{x}}$ with $\mathbf{v} = (v_x, 0, 0)$

$$\begin{aligned} E'_x &= E_x, & E'_y &= \gamma(E_y - v_x B_z), & E'_z &= \gamma(E_z + v_x B_y), \\ B'_x &= B_x, & B'_y &= \gamma(B_y + v_x E_z/c^2), & B'_z &= \gamma(B_z - v_x E_y/c^2). \end{aligned}$$

(b) Field tensor from $\mathbf{A}$, Maxwell covariant form

Definitions: 4-potential $\mathbf{A} = (\phi/c, \mathbf{A})$ and 4-wave vector $\mathbf{K} = (\omega/c, \mathbf{k})$.

For propagation along $\hat{\mathbf{z}}$ with $K^\mu = (\omega/c, 0, 0, k)$. Compute the phase $K^\mu X_\mu$ in $(+, -, -, -)$:

$$K^\mu X_\mu = (\omega/c)(ct) - kz = \omega t - kz.$$

Substitute into A_y :

$$A_y = -\frac{\sin(\omega t - kz)}{\sqrt{\omega k}}.$$

Differentiate w.r.t. t :

$$\partial_t A_y = -\frac{\omega \cos(\omega t - kz)}{\sqrt{\omega k}}.$$

Differentiate w.r.t. z :

$$\partial_z A_y = \frac{k \cos(\omega t - kz)}{\sqrt{\omega k}}.$$

Read off E_y from $\mathbf{E} = -\partial_t \mathbf{A} - \nabla \phi$:

$$E_y = -\partial_t A_y = \frac{\omega \cos(\omega t - kz)}{\sqrt{\omega k}} = \sqrt{\omega/k} \cos(\omega t - kz).$$

Read off B_x from $\mathbf{B} = \nabla \times \mathbf{A}$ (only A_y non-zero, A_y depends on z, t):

$$B_x = -\partial_z A_y = -\sqrt{k/\omega} \cos(\omega t - kz).$$

Compute $F^{02} = \partial^0 A^2 - \partial^2 A^0$ with $\partial^0 = (1/c)\partial_t$, $A^0 = 0$, $A^2 = A_y$:

$$F^{02} = (1/c)\partial_t A_y.$$

Substitute $\partial_t A_y = -E_y$:

$$F^{02} = -E_y/c.$$

Compute $F^{32} = \partial^3 A^2 - \partial^2 A^3$ with $\partial^3 = -\partial_z$, $A^3 = 0$:

$$F^{32} = -\partial_z A_y = B_x.$$

Field-tensor entries (independent non-zero ones, others fixed by antisymmetry):

$$\boxed{F^{02} = -E_y/c, F^{32} = B_x; \quad E_y = \sqrt{\omega/k} \cos(\omega t - kz), B_x = -\sqrt{k/\omega} \cos(\omega t - kz)}.$$

4-current and continuity. Define $J^\nu = (\rho c, \mathbf{j})$. 3-vector continuity:

$$\partial_t \rho + \nabla \cdot \mathbf{j} = 0.$$

4-vector form:

$$\boxed{\partial_\nu J^\nu = 0}.$$

Maxwell covariant form. Compute $\partial_\mu F^{\mu 0}$ using $F^{00} = 0$ and $F^{i0} = E^i/c$:

$$\partial_\mu F^{\mu 0} = \partial_i F^{i0} = \nabla \cdot \mathbf{E}/c.$$

Apply Gauss's law $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$:

$$\partial_\mu F^{\mu 0} = \rho/(c\epsilon_0).$$

Use $\mu_0 \epsilon_0 = 1/c^2$, i.e. $1/(c\epsilon_0) = \mu_0 c$:

$$\partial_\mu F^{\mu 0} = \mu_0 c \rho = \mu_0 J^0.$$

Compute $\partial_\mu F^{\mu i}$ using $F^{0i} = -E^i/c$ and $F^{ji} = -\epsilon^{jik} B_k$:

$$\partial_\mu F^{\mu i} = (1/c)\partial_t(-E^i/c) + (\nabla \times \mathbf{B})^i = -\partial_t E^i/c^2 + (\nabla \times \mathbf{B})^i.$$

Apply Ampère–Maxwell $\nabla \times \mathbf{B} - (1/c^2)\partial_t \mathbf{E} = \mu_0 \mathbf{j}$:

$$\partial_\mu F^{\mu i} = \mu_0 j^i = \mu_0 J^i.$$

Combine both cases:

$$\boxed{\partial_\mu F^{\mu \nu} = \mu_0 J^\nu}.$$

Physical meaning of $f_0 = J^\nu F_{\nu 0}$. Use $F_{00} = 0$ and $F_{i0} = -F^{i0} = -E_i/c$:

$$f_0 = J^0 F_{00} + J^i F_{i0} = j^i \cdot (-E_i/c) = -\mathbf{j} \cdot \mathbf{E}/c.$$

Multiply by c :

$$cf_0 = -\mathbf{j} \cdot \mathbf{E}.$$

Interpretation: $\mathbf{j} \cdot \mathbf{E}$ is the power per unit volume delivered by the field to the charges (Ohmic dissipation / energy transfer), so cf_0 is minus the rate at which the field does work on matter per unit volume.

(c) **Pauli–Lubanski** $U_W \perp P$

Definition:

$$W^\mu = (\mathbf{s} \cdot \mathbf{p}, sE/c), \quad U_W^\mu = W^\mu / (m_0 c |\mathbf{s}|).$$

Compute $\mathbf{W} \cdot \mathbf{P}$ with $\mathbf{P} = (E/c, \mathbf{p})$:

$$\mathbf{W} \cdot \mathbf{P} = W^0 P^0 - \mathbf{W} \cdot \mathbf{p}.$$

Substitute $W^0 = \mathbf{s} \cdot \mathbf{p}$ and $\mathbf{W} = sE/c$:

$$\mathbf{W} \cdot \mathbf{P} = (\mathbf{s} \cdot \mathbf{p})(E/c) - (sE/c) \cdot \mathbf{p}.$$

Factor common factors:

$$\mathbf{W} \cdot \mathbf{P} = (\mathbf{s} \cdot \mathbf{p})(E/c) - (\mathbf{s} \cdot \mathbf{p})(E/c) = 0.$$

Divide by $m_0 c |\mathbf{s}|$:

$$\boxed{U_W \cdot P = 0}.$$

(d) **Wire fields in three frames; can two wires interact purely electrically?**

Lab frame S. Wire along $\hat{\mathbf{x}}$, $\mathbf{J} = j_x \hat{\mathbf{x}}$ with $J^0 = 0$ (neutral wire). Apply Gauss's law (no net charge) and Ampère's law:

$$\boxed{\mathbf{E}_S = \mathbf{0}, \quad \mathbf{B}_S = \frac{\mu_0 I}{2\pi r} \hat{\phi}}.$$

At a point at distance r above the wire (along $+\hat{\mathbf{y}}$), the field is $\mathbf{B}_S = B\hat{\mathbf{z}}$ with $B = \mu_0 I / (2\pi r)$.

Frame moving perpendicular to wire, $\mathbf{u} = (0, v_0, 0)$. Apply Lorentz boost along $\hat{\mathbf{y}}$ with $\gamma = 1/\sqrt{1 - v_0^2/c^2}$. Compute E'_x :

$$E'_x = \gamma(E_x + v_0 B_z) = \gamma v_0 B.$$

Compute E'_z :

$$E'_z = \gamma(E_z - v_0 B_x) = 0.$$

Compute B'_z :

$$B'_z = \gamma(B_z + v_0 E_x/c^2) = \gamma B.$$

Compute B'_x :

$$B'_x = \gamma(B_x - v_0 E_z/c^2) = 0.$$

Both $\mathbf{E}'$ and $\mathbf{B}'$ are non-zero. The wire remains neutral (perpendicular boost preserves linear charge density along the wire); the new $\mathbf{E}'$ arises from Lorentz transformation of $\mathbf{B}$.

Frame co-moving with electrons, $\mathbf{u} = (v_0, 0, 0)$. Apply Lorentz boost along $\hat{\mathbf{x}}$. Compute E'_y :

$$E'_y = \gamma(E_y - v_0 B_z) = -\gamma v_0 B = -\frac{\gamma v_0 \mu_0 I}{2\pi r}.$$

Compute B'_z :

$$B'_z = \gamma(B_z - v_0 E_y/c^2) = \gamma B.$$

Other components vanish:

$$E'_x = 0, \quad E'_z = 0; \quad B'_x = 0, \quad B'_y = 0.$$

Physically: in this frame the electrons are stationary while positive ions are Lorentz-contracted, so the wire appears positively charged, giving a radial $\mathbf{E}'$; the residual current of ions produces $\mathbf{B}'$.

Two parallel wires, separation d . For the inter-wire force to be purely electrical we need $\mathbf{B}' = \mathbf{0}$ at the second wire in some frame.

Compute field invariants in lab S:

$$I_1 = \mathbf{E} \cdot \mathbf{B} = 0, \quad I_2 = E^2 - c^2 B^2 = -c^2 B^2 < 0.$$

Invariants are frame-independent, so in any frame S'' :

$$I_2 = E''^2 - c^2 B''^2 < 0.$$

Rearrange:

$$c^2 B''^2 > E''^2 \geq 0.$$

Hence $B'' \neq 0$ in every frame. The inter-wire force cannot be purely electrical:

No frame makes the inter-wire force purely electrical	□
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(e) Inverse Compton: maximum photon energy

Head-on geometry: photon $\mathbf{K} = (\omega/c, k_x, 0, 0)$ with $k_x = -\omega/c$ (incoming along $-\hat{\mathbf{x}}$); electron $\mathbf{P} = (W/c, p, 0, 0)$ moving along $+\hat{\mathbf{x}}$ with $W \gg M_0 c^2$ so $\gamma = W/(M_0 c^2) \gg 1$ and $\beta \approx 1$.

Step 1: boost to electron rest frame. Apply Lorentz transformation to K^0 with velocity $+v\hat{\mathbf{x}}$:

$$\omega^*/c = \gamma(K^0 - \beta K^1) = \gamma(\omega/c - \beta(-\omega/c)).$$

Factor ω/c :

$$\omega^* = \gamma\omega(1 + \beta).$$

Take ultra-relativistic limit $\beta \rightarrow 1$:

$$\omega^* \approx 2\gamma\omega.$$

Step 2: scattering in rest frame. For maximum lab energy, the photon back-scatters ($\theta^* = \pi$). Apply Compton formula:

$$\frac{1}{\hbar\omega^{*'}} - \frac{1}{\hbar\omega^*} = \frac{1 - \cos \pi}{M_0 c^2} = \frac{2}{M_0 c^2}.$$

Apply Thomson limit $\hbar\omega^* \ll M_0 c^2$:

$$\omega^{*'} \approx \omega^*.$$

Step 3: boost back to lab. The scattered photon now moves along $+\hat{\mathbf{x}}$ (same direction as electron motion). Apply inverse boost (lab moves at $-v\hat{\mathbf{x}}$ relative to rest frame):

$$\omega'_{\text{lab}} = \gamma\omega^{*'}(1 + \beta) \approx 2\gamma\omega^{*'}.$$

Substitute $\omega^{*'} \approx 2\gamma\omega$:

$$\omega'_{\text{lab}} \approx 4\gamma^2\omega.$$

Multiply by $\hbar$ to convert to photon energy:

$$E_{\text{ph}}^{\text{max}} = \hbar\omega'_{\text{lab}} = 4\gamma^2\hbar\omega = 4\gamma^2 E_{\text{ph}}.$$

Substitute $\gamma = W/(M_0 c^2)$:

$E_{\text{ph}}^{\text{max}} = 4 \left(\frac{W}{M_0 c^2} \right)^2 E_{\text{ph}}.$	□
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